```
!pip install transformers
!pip install sentencepiece
!pip install torch
!pip install pandas
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transform
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->h
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->h
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
```

```
from transformers import pipeline
import pandas as pd
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

class CustomTranslator:
    def __init__(self, model_name):
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
        self.model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

    def __call__(self, text):
        inputs = self.tokenizer(text, return_tensors="pt", padding=True)
        # A default max_length is used, it can be adjusted if needed.
        outputs = self.model.generate(**inputs, max_length=512)
        translated_text = self.tokenizer.decode(outputs[0], skip_special_tokens=True)
        return [{"translation_text": translated_text}]

translator_fr = CustomTranslator("Helsinki-NLP/opus-mt-en-fr")
translator_de = CustomTranslator("Helsinki-NLP/opus-mt-en-de")
translator_hi = CustomTranslator("Helsinki-NLP/opus-mt-en-hi")

print("Models Loaded Successfully")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

```
config.json:        1.42k/? [00:00<00:00, 120kB/s]

tokenizer_config.json: 100%                                    42.0/42.0 [00:00<00:00, 4.09kB/s]

source.spm: 100%                                               778k/778k [00:00<00:00, 10.5MB/s]

target.spm: 100%                                               802k/802k [00:00<00:00, 41.3MB/s]

vocab.json:         1.34M/? [00:00<00:00, 35.8MB/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip
  warnings.warn("Recommended: pip install sacremoses.")
```

```
pytorch_model.bin: 100%                                        301M/301M [00:02<00:00, 232MB/s]

Loading weights: 100%                                          258/258 [00:00<00:00, 442.67it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                                        301M/301M [00:06<00:00, 62.4MB/s]
```

```
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
```

```
generation_config.json: 100%                                   293/293 [00:00<00:00, 9.52kB/s]

config.json:        1.33k/? [00:00<00:00, 37.6kB/s]

tokenizer_config.json: 100%                                    42.0/42.0 [00:00<00:00, 946B/s]

source.spm: 100%                                               768k/768k [00:00<00:00, 6.23MB/s]

target.spm: 100%                                               797k/797k [00:00<00:00, 25.9MB/s]

vocab.json:         1.27M/? [00:00<00:00, 17.4MB/s]

pytorch_model.bin: 100%                                        298M/298M [00:06<00:00, 44.5MB/s]

Loading weights: 100%                                          258/258 [00:00<00:00, 550.26it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                                        298M/298M [00:05<00:00, 62.4MB/s]
```

```
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
```

```
generation_config.json: 100%                                   293/293 [00:00<00:00, 14.9kB/s]

config.json:        1.39k/? [00:00<00:00, 54.9kB/s]

tokenizer_config.json: 100%                                    44.0/44.0 [00:00<00:00, 2.28kB/s]

source.spm: 100%                                               812k/812k [00:00<00:00, 11.5MB/s]

target.spm: 100%                                               1.07M/1.07M [00:00<00:00, 43.6MB/s]

vocab.json:         2.10M/? [00:00<00:00, 50.1MB/s]

pytorch_model.bin: 100%                                        306M/306M [00:03<00:00, 160MB/s]

Loading weights: 100%                                          258/258 [00:00<00:00, 476.43it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                                        306M/306M [00:04<00:00, 96.2MB/s]
```

```
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
```

```
generation_config.json: 100%                                   293/293 [00:00<00:00, 16.4kB/s]

Models Loaded Successfully
```

```
df = pd.read_csv("/content/eng_-french.csv")
df
```

| | English words/sentences | French words/sentences |
|---|---|---|
| **0** | Hi. | Salut! |
| **1** | Run! | Cours ! |
| **2** | Run! | Courez ! |
| **3** | Who? | Qui ? |
| **4** | Wow! | Ça alors ! |
| **...** | ... | ... |
| **175616** | Top-down economics never works, said Obama. "T... | « L'économie en partant du haut vers le bas, ç... |
| **175617** | A carbon footprint is the amount of carbon dio... | Une empreinte carbone est la somme de pollutio... |
| **175618** | Death is something that we're often discourage... | La mort est une chose qu'on nous décourage sou... |
| **175619** | Since there are usually multiple websites on a... | Puisqu'il y a de multiples sites web sur chaqu... |
| **175620** | If someone who doesn't know your background sa... | Si quelqu'un qui ne connaît pas vos antécédent... |

175621 rows × 2 columns

```
sentences = df["English words/sentences"].tolist()

print(sentences[:5])
```

```
['Hi.', 'Run!', 'Run!', 'Who?', 'Wow!']
```

```
french_translations = []

for sentence in sentences:
    translated = translator_fr(sentence)[0]['translation_text']
    french_translations.append(translated)

df["French"] = french_translations

df.head()
```

```
german_translations = []
hindi_translations = []

for sentence in sentences:
    german = translator_de(sentence)[0]['translation_text']
    hindi = translator_hi(sentence)[0]['translation_text']

    german_translations.append(german)
    hindi_translations.append(hindi)

df["German"] = german_translations
df["Hindi"] = hindi_translations

df
```

```
df.to_csv("translated_output.csv", index=False)

print("File Saved Successfully")
```

```
File Saved Successfully
```

```
from google.colab import files
files.download("translated_output.csv")
```

```
complex_sentences = [
    "Although the economy is growing, unemployment remains a concern.",
    "When neural networks are trained properly, they perform better in prediction tasks.",
    "If the dataset is clean, the model will produce accurate results.",
    "Did you complete the assignment for NLP class?",
    "The transformer model uses attention mechanisms for translation."
]

for sentence in complex_sentences:
    print("English:", sentence)
    print("French:", translator_fr(sentence)[0]['translation_text'])
    print("German:", translator_de(sentence)[0]['translation_text'])
    print("Hindi:", translator_hi(sentence)[0]['translation_text'])
    print()
```

```
English: Although the economy is growing, unemployment remains a concern.
French: Bien que l'économie se développe, le chômage demeure un sujet de préoccupation.
German: Obwohl die Wirtschaft wächst, bleibt die Arbeitslosigkeit besorgniserregend.
Hindi: हालाँकि बेरोज़गारी बढ़ती जा रही है, फिर भी बेरोज़गारी की चिंता बढ़ती जा रही है ।

English: When neural networks are trained properly, they perform better in prediction tasks.
French: Lorsque les réseaux neuraux sont correctement formés, ils effectuent mieux dans les tâches de prédiction.
German: Wenn neuronale Netzwerke richtig ausgebildet werden, führen sie bessere Vorhersageaufgaben durch.
Hindi: जब तंत्रिका नेटवर्क को सही तरीके से प्रशिक्षित किया जाता है, तो वे भविष्यवाणी के कार्यों में बेहतर कार्य करते हैं ।

English: If the dataset is clean, the model will produce accurate results.
French: Si l'ensemble de données est propre, le modèle produira des résultats précis.
German: Wenn der Datensatz sauber ist, liefert das Modell genaue Ergebnisse.
Hindi: अगर डेटासेट साफ है, तो मॉडल सही नतीजे पर पहुँचा देगा ।

English: Did you complete the assignment for NLP class?
French: Avez-vous terminé l'affectation pour la classe NLP?
German: Hast du den Auftrag für den NLP-Kurs abgeschlossen?
Hindi: क्या आपने एनएलपी क्लास के लिए यह काम पूरा किया?

English: The transformer model uses attention mechanisms for translation.
French: Le modèle de transformateur utilise des mécanismes d'attention pour la traduction.
```